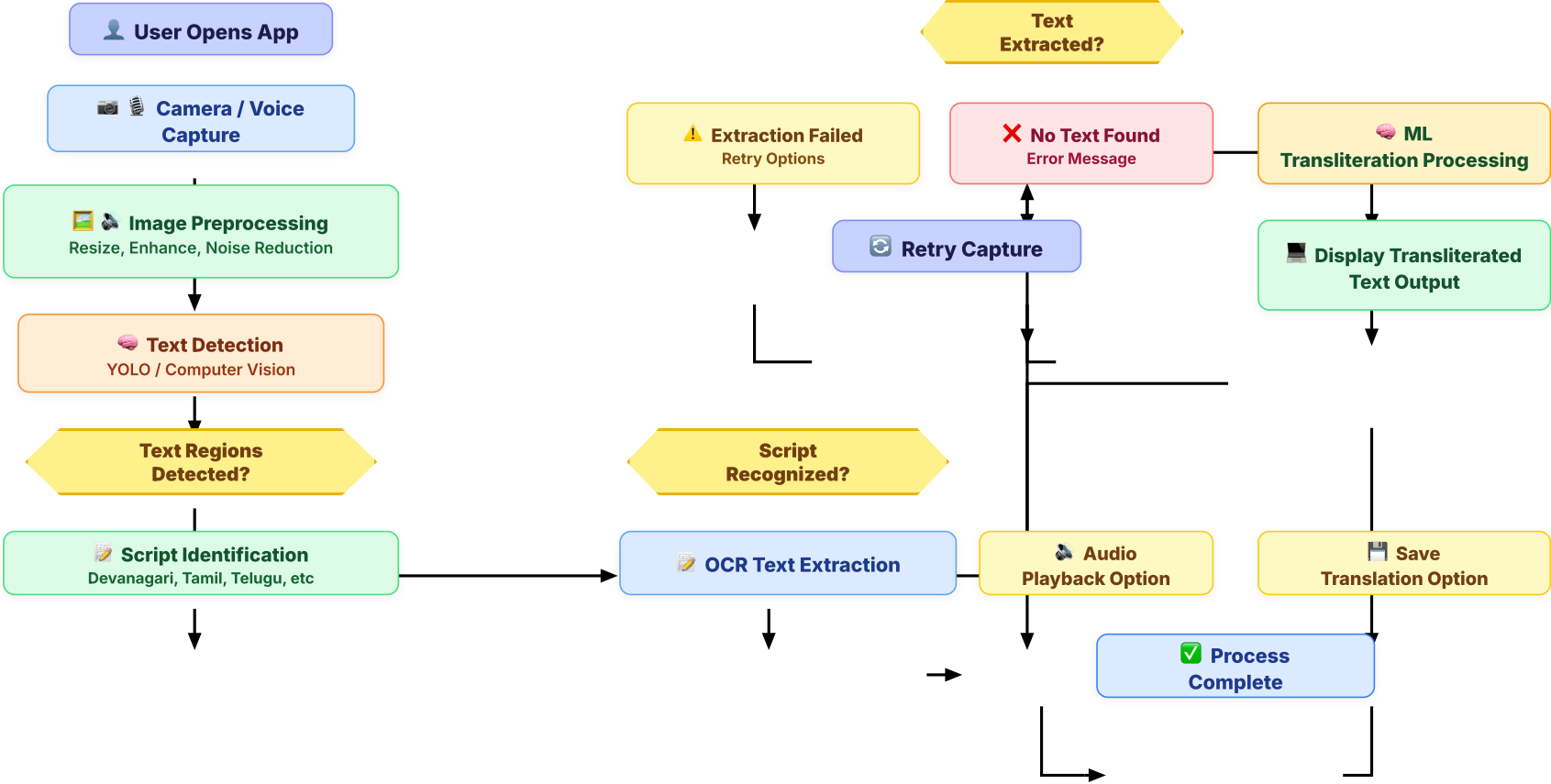


# TECHNICAL APPROACH

## Technology Stack

- **Mobile App:** React 19, Vite, Capacitor Native Android (Universal APK)
- **Backend / Architecture:** 100% Offline Edge Service Worker, CacheStorage
- **AI/ML & Voice Tools:**
  - **Speech / Audio:** IndicWav2Vec Acoustic Model, Web Audio Formant Synthesizer
  - **Translation:** AI4Bharat IndicTrans2 (INT8 ONNX), Jim Cummins Dual-Iceberg
- **Data Processing:** Pandas, NumPy, CIIL 12,000+ Tribal Lexicon, NIPUN Matrix
- **Target Languages:** Santhali (OI Chiki), Ho (Warang Chiti), Mundari, Kudmali, Sadri
- **Classroom Hardware:** Low-cost 2GB Android Tablets, e-Vidyavahini 2.0 Edge
- **Database & Storage:** IndexedDB, SQLite Offline Cache / Firebase Sync
- **Security & Privacy:** 100% On-Device DPDP Act 2023 Isolation, Keystore v2+v3
- **Pedagogical Standard:** NEP 2020 MTB-MLE, NCF-FS & NIPUN Bharat 80:20
- **Output Modalities:** Native Script Display, Offline TTS, 300 DPI Audio QR Print





Firebase



NumPy



OCR



Power BI



Python



React



GitHub



LinkedIn



Google Cloud



AWS



AICTE



AI4Bharat



MongoDB



Node.js



OpenAI

### Input Stage

Signboard / Textbook  
(Any Script) → Camera OCR



Optical character recognition

### Processing Stage

Script Conversion Engine

- Input Script Analysis
- Character Mapping & Transliteration
- Target Script Synthesis

### Output Stage

Output in Chosen Script



User Selected Language / Alphabet

# SARJOM Technical Approach: Engineering Deep-Dive

SIH 2026 Problem SIH26042 · Client: Govt. of Jharkhand (DHTE) · Team NextGen Innovators

100% OFFLINE EDGE PROTOTYPE

## 1. Input & Acoustic DSP Processing

**Overcoming Tin-Roof Monsoon Noise:** Jharkhand forest schools experience heavy ambient noise (75–82 dB) on corrugated metal classroom roofs.

- **Web Audio DSP Noise Gate:** Biquad bandpass filter restricts incoming frequencies strictly to human vocal fundamentals (300 Hz – 3,400 Hz).
- **16 kHz PCM Acoustic Stream:** Operates directly on low-cost 2GB Android tablets with zero cloud speech roundtrips.
- **Echo Cancellation (AEC):** Tablet speaker broadcasts native pronunciations without re-triggering microphone feedback loops.
- **Tactile Fallback:** If acoustic SNR < 12 dB, teachers tap 1-touch prompt chips to bypass speech without stopping instruction.

## 2. Edge AI & Munda Morphology Engine

**Beyond Shallow Word Lookup:** Austroasiatic Munda tongues (Santhali, Ho, Mundari) are polysynthetic and agglutinative.

- **AI4Bharat IndicTrans2 INT8:** Dynamically quantized transformer runtime executes in browser memory in ~34 MB of RAM (sub-50ms latency).
- **Dual-Iceberg Cognitive Bridge:** Maps BICS (basic conversational fluency) into CALP (cognitive academic language proficiency) in Hindi.
- **80:20 Scaffolding:** Balvatika starts at 80% tribal mother tongue; gradually inverts to 80% standard Hindi by Grade 3 per NIPUN Bharat.
- **Unicode 5.1 Matrix:** Authentic glyph mappings for OI Chiki (U+1C50), Warang Chiti (U+118A0), and Devanagari.

## 3. Multi-Modal Output & Take-Home Audio

**Closing the Forest Hamlet Homework Gap:** Village parents are often non-literate and lack internet connectivity.

- **Offline Formant Speech Synthesis:** Emulates human vocal tract formant resonance (F1: 620Hz, F2: 1720Hz) directly via Web Audio API.
- **Dynamic Audio QR Worksheets:** Reed-Solomon Error Correction Level M vector QR codes printed at 300 DPI encode lesson audio.
- **Universal Android APK:** V1, V2, and V3 APK signed, installable on any low-cost Android 9+ tablet or phone.
- **e-Vidyavahini 2.0 Sneakernet Sync:** Formative scores queue in IndexedDB for weekly MicroSD/USB block synchronization.

### 30-Second Evaluator Presentation Script:

"Respected Jury, our Technical Approach is engineered specifically for remote tribal schools in Jharkhand: **zero internet, 2GB Android tablets, and 80-decibel monsoon classroom noise**. As demonstrated in our system flowchart, the teacher speaks in Hindi or inputs text. Our **Web Audio DSP Noise Gate** instantly filters tin-roof rumble. Our **AI4Bharat IndicTrans2 INT8 engine** runs 100% on-device inside client memory in under 35 megabytes of RAM—requiring **zero cloud API calls**. It translates instructions into authentic Santhali OI Chiki, Ho Warang Chiti, and Mundari using Jim Cummins' Dual-Iceberg scaffolding. The app speaks the phrase aloud offline using our **formant synthesizer**, displays authentic native glyphs, and generates **take-home Audio QR worksheets** for non-literate village parents. It is packaged into a universal Android APK and complies 100% with the DPDP Act 2023."